

Gregory Dalbey

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Senior Software Engineer

Senior Software Engineer with 30+ years of professional software development experience, including deep ownership of architecture, modernization, and delivery in high-impact systems. Over the last decade, focused on replacing fragile legacy applications with robust, maintainable platforms that improve business velocity and reduce risk. Known for pragmatic design, operational reliability, and close partnership with stakeholders across engineering and business teams.

Highlights

- Leads modernization of business-critical systems with careful continuity planning
- Designs and delivers full-stack solutions from schema to services to UI
- Strong track record in performance, automation, and maintainability improvements
- Bridges technical strategy and day-to-day delivery for high-dependency workflows
- Active in modern development practices, including AI-assisted and agentic workflows

Core Technical Skills

- **Architecture & Delivery:** system design, migration planning, legacy refactoring, requirements translation
- **Backend:** .NET 6/.NET Core, ASP.NET MVC, service design, batch processing, integration pipelines
- **Frontend:** Blazor, React (working knowledge), jQuery
- **Data:** SQL Server, Azure SQL, schema design, data workflow modernization
- **Cloud & Security:** Azure Blob Storage, Azure Key Vault, secure configuration/secrets patterns

Professional Experience

Senior Software Engineer | Musicnotes | Madison, WI

Jun 2016 – Present

- Leading a multi-phase modernization of a mission-critical Access-based publishing workflow ("Push Process") that processes approximately 1,500 products weekly and powers the majority of product availability on the ecommerce site.
- Driving architecture and migration strategy for lift-and-shift transformation from multiple entangled legacy systems to a cohesive platform connecting catalog metadata, digital ownership records, and royalties-related downstream dependencies.
- Partnering with a lean operations footprint (2-person process ownership team) to preserve continuity in a high-throughput workflow while reducing long-term maintenance burden.
- Implementing a pluggable configuration abstraction and Azure Key Vault integration to remove sensitive keys/connection strings from application config and support secure runtime retrieval for integrations and data stores.
- Supporting AI-safe engineering practices by reducing sensitive configuration exposure in codebases used with AI-assisted and agentic development workflows.

- Architected and implemented a royalties reporting batch processor generating 5,000+ raw/formatted reports quarterly.
- Reduced royalties reporting runtime from 24+ hours to under 2 hours per cycle.
- Automated publisher-facing report formatting, cutting manual effort from approximately 7 days per quarter to hours.
- Designed and implemented a queued messaging processor integrated with Adobe ACS, supporting 1,000,000+ annual transactional/marketing emails.
- Modernized Hal Leonard purchasing workflows in .NET 6, removing obsolete dependencies and unlocking broader API usage.
- Delivered key customer-facing capabilities including Digital Music Books, Pro Membership, Educator Discount, and Wish List.

Selected Initiatives (Musicnotes)

Push Process Modernization (2024-Present)

- Leading rewrite strategy for legacy Access publishing process and related data flows.
- Defining migration boundaries/phasing across upstream and downstream systems to reduce refactor risk.
- Establishing design direction for a unified, maintainable platform architecture.

Secrets Modernization / Azure Key Vault (2024-Present)

- Introduced interface-driven configuration design to decouple legacy configuration manager dependencies.
- Integrated secure secret retrieval for DB connections, email systems, and third-party APIs.
- Improved codebase readiness for AI-assisted development by reducing sensitive-data exposure.

Digital Music Books Publishing Platform (2022)

- Built a production prototype for DMB publishing, including schema design, Blazor application development, and backend publishing services.
- Implemented publishing flow for metadata and digital assets (book files, thumbnails) to ecommerce systems.
- Added Interactive DMB support by publishing interactive assets to Azure Blob Storage for post-purchase customer access.

Software Engineer | Sonic Foundry | Madison, WI

Jan 2015 – Jun 2016

- Implemented accessibility features for media players used in educational and enterprise video environments.
- Standardized keyboard control behavior across player variants to improve consistency/usability.
- Collaborated with QA/support teams on customer-prioritized fixes and release quality.

Software Developer Consultant | Yahara Software | Madison, WI

Dec 2012 – Jan 2015

- Built and modernized client applications using ASP.NET MVC, jQuery, SQL Server, WPF, and SignalR.
- Developed distributed online livestock auction functionality and real-time interaction workflows.
- Integrated specialized medical automation systems with modern control/process interfaces.

Applications Developer | UW Facilities Planning & Management | Madison, WI

Jan 2006 – Dec 2011

- Redesigned time-tracking software serving 2,000+ university employees.
- Maintained and enhanced university-wide parking registration systems.
- Migrated and improved departmental registration/portal applications.

Additional Earlier Experience

Earlier professional work includes software development and system design across varied business environments, forming a strong foundation for long-term architecture and modernization leadership.

Education

Bachelor of Science in Computer Science, Minor in Music North Central University, Naperville, IL (1995)

Professional Development

- Ongoing self-directed study in Kubernetes and Docker with local cluster experimentation.
- Active use of AI-assisted and agentic development techniques to improve delivery speed and software quality.